

OS

Definition

An operating system (OS) is the software layer that acts as an interface between user applications and the computer hardware. Its primary functions include:

- **Resource Management:** Allocates CPU time, memory, and storage to various processes.
- **Process Management:** Creates, schedules, and terminates processes to ensure efficient multitasking.
- **Memory Management:** Controls access to RAM, ensuring that each process operates within its own isolated address space.
- **File System Management:** Organizes and manages data stored on disks, providing a structure for file access.
- **I/O Device Control:** Provides a unified interface for applications to interact with hardware devices like keyboards, monitors, and network adapters.
- **Security & Protection:** Regulates user access through authentication and guards against unauthorized system interaction.

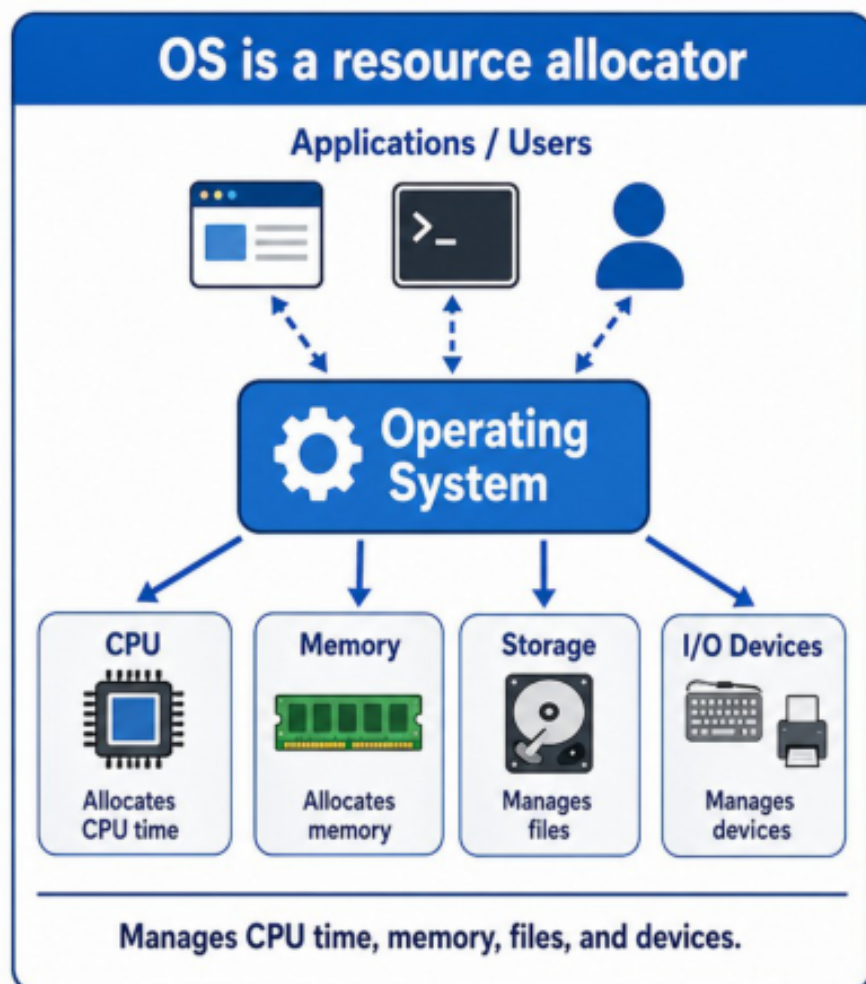


Figure 1: OS is a resource allocator

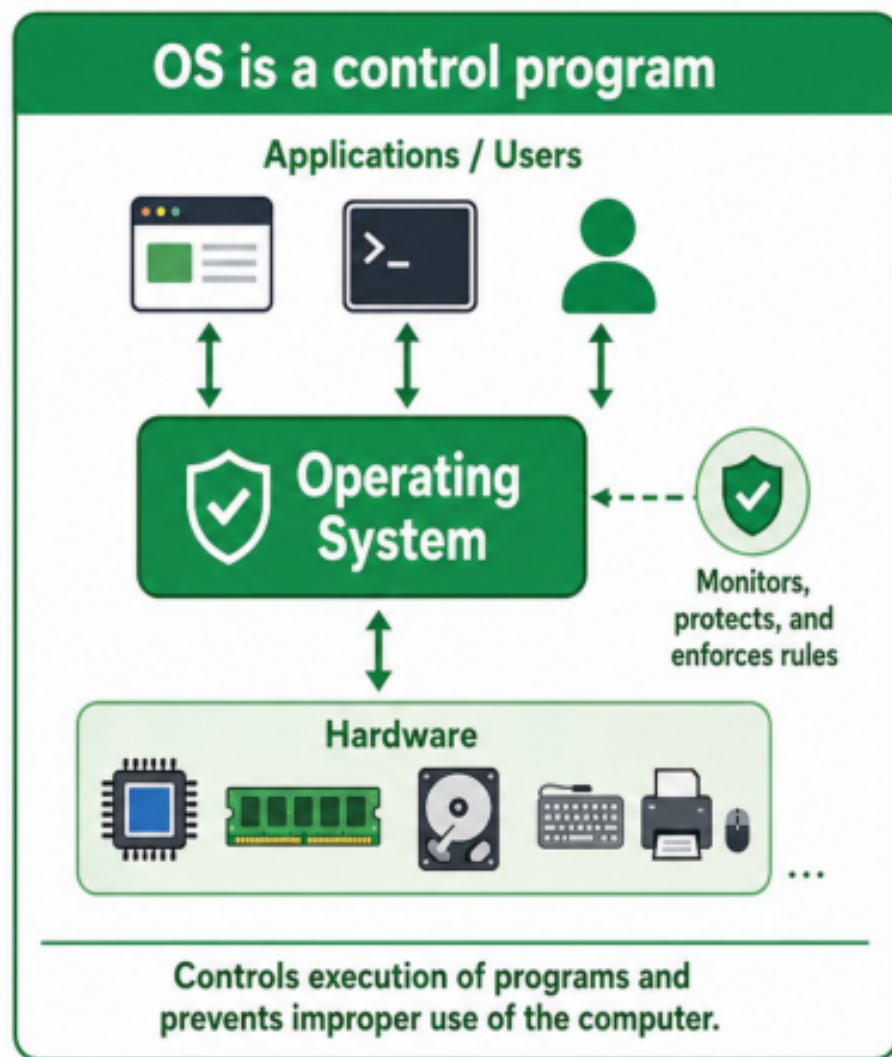


Figure 2: OS is a control program

OSTEP defines it in three steps which are:

- Virtualization
- Concurrency
- Parallelism

Computer System Structure

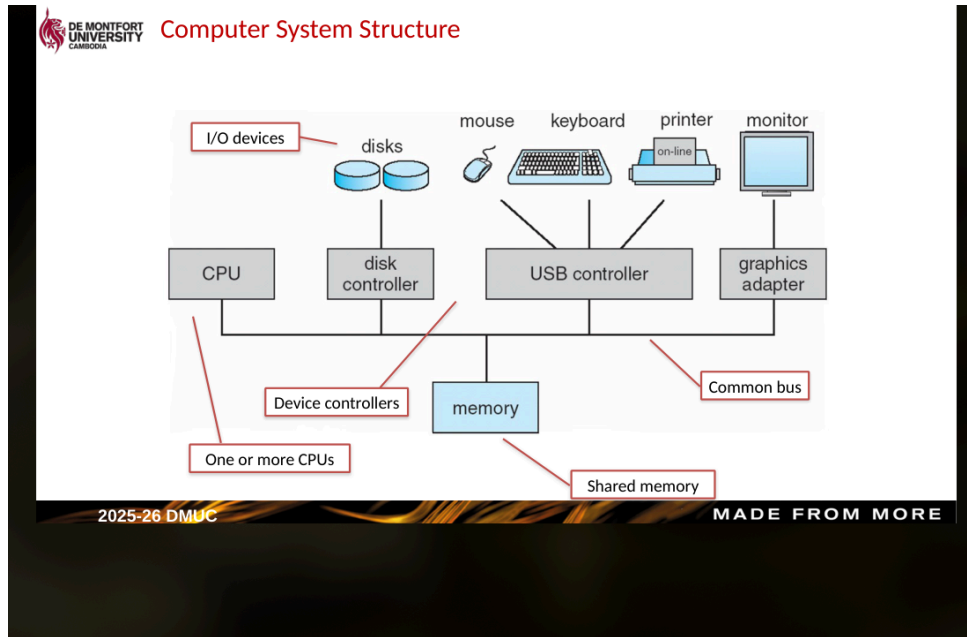


Figure 3: Computer System Structure

Interrupts

- OS is interrupt driven - an interrupt is a signal sent to the OS:
 - Hardware interrupt by one of the devices
 - Software interrupt generated by user request
 - Trap or exception generated by an error

Dual-Mode Operation

- Dual-mode operation allows the OS to protect itself, as well as other components.
 - User mode and kernel mode

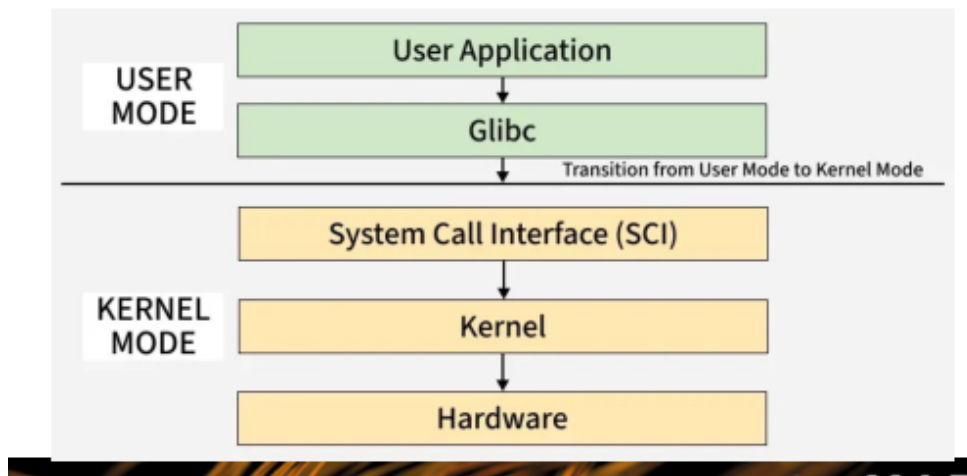


Figure 4: User mode and Kernel mode

Kernel Mode

- In kernel mode, the operating system has unrestricted access to the underlying hardware and memory.
 - Can execute privileged instructions that are not available to user programs.

- Can manage system resources such as CPU scheduling, memory allocation, and I/O devices.
- Can handle interrupts, exceptions, and traps directly.
- Executes with full authority to modify the system state, including kernel data structures.

User Mode

- In user mode, the operating system restricts the program's access to hardware and system resources to prevent system crashes or malicious activity.
 - Programs have limited access to memory and cannot directly access I/O devices.
 - Privileged instructions are prohibited; any attempt to execute them triggers a trap or exception that the kernel must handle.
 - Programs must request system services via system calls to perform restricted operations.
 - The mode provides a protected environment where applications can run safely without interfering with the kernel or other running processes.

Comparison

Feature	User Mode	Kernel Mode
Access	Restricted	Unrestricted
Instructions	Non-privileged only	Privileged and non-privileged
Memory	Isolated address space	Full system memory
Services	Request via system calls	Direct access to hardware
Stability	Safe/Protected	Critical/Requires care